

Skill Arena - Phase 6 Document 1

Multi-Game Framework & Game SDK Architecture

Document Purpose

Define how future games integrate into Skill Arena without requiring major platform redesign.

Business Objective

Transform Skill Arena from a single-game platform into a scalable multi-game competitive ecosystem.

Platform Vision

Every future game should inherit wallets, tournaments, matchmaking, leaderboards, replays, security and progression systems automatically.

Multi-Game Framework

Central framework allowing multiple game types to coexist within the same platform architecture.

Game Integration Philosophy

Build once, integrate many. Core platform services reused across all games.

Game SDK Overview

Developer toolkit used to connect new games to platform services.

SDK Responsibilities

Authentication, Matchmaking, Wallet Integration, Replay Integration, Events and Analytics.

Supported Game Types

Maze Games, Puzzle Games, Reaction Games, Logic Games, Racing Challenges and Future Skill-Based Games.

Game Registration System

Each game receives a unique identifier, configuration profile and integration package.

Shared Platform Services

Wallets, Treasury, Profiles, Achievements, Seasons, Tournaments and Security.

Game Metadata Model

Game ID, Version, Ruleset, Difficulty Model, Reward Model and Validation Logic.

Challenge Integration

Games publish challenge definitions and progression structures to the platform.

Replay Compatibility

Every game must support replay generation and replay verification.

Leaderboard Compatibility

Games integrate into seasonal, lifetime and game-specific rankings.

Tournament Compatibility

Games may participate in tournaments using common tournament infrastructure.

Reward Compatibility

Rewards distributed through centralized wallet and treasury services.

Analytics Integration

Gameplay metrics automatically flow into platform analytics systems.

Security Requirements

All games must operate under server-authoritative validation rules.

Game Certification Process

New games must pass security, replay, validation and integration testing.

Developer Framework

Provide APIs, SDKs, documentation and integration standards for future development.

Version Management

Support multiple game versions and controlled release strategies.

Scalability Requirements

Framework supports dozens or hundreds of games over time.

Monitoring Requirements

Track game health, performance, engagement and operational metrics.

Audit Requirements

Game integrations, updates and releases permanently recorded.

Acceptance Criteria

New games can be integrated rapidly while inheriting core platform capabilities.